

# Design Docs

17-313 Fall 2026

Foundations of Software Engineering

<https://cmu-17313q.github.io>

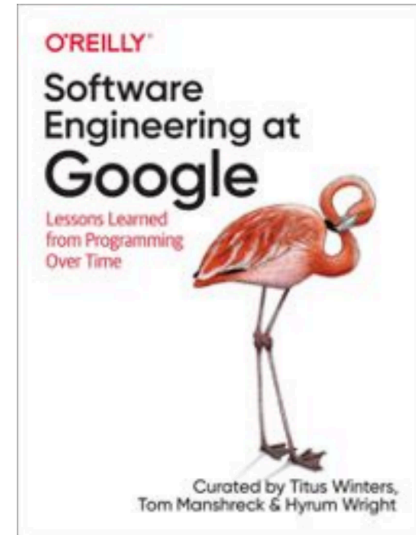
Eduardo Feo-Flushing

# Administrivia

- Team surveys. The first one is due today!
- Problem Set 1, P2A
- Quiz 1: Next Sunday, September 20

# Types of documentation

- Reference documentation (incl. code comments)
- Design documents
- Tutorials
- Conceptual documentation
- Landing pages



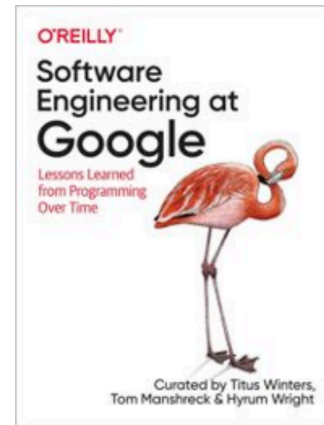
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## NOTES

These categories aren't mutually exclusive; most real documentation mixes several of them. Today we're focusing specifically on design documents: the kind you write before any code exists.

# Design documents

- Code review before there is code!
- Collaborative (Google Docs)
- Ensure various concerns are covered, such as:  
security implications, internationalization, storage requirements, and privacy concerns.
- A good design doc should cover
  - Goals and use cases for the design
  - Implementation ideas (not too specific!)
  - Propose key design decisions with an emphasis on their individual tradeoffs



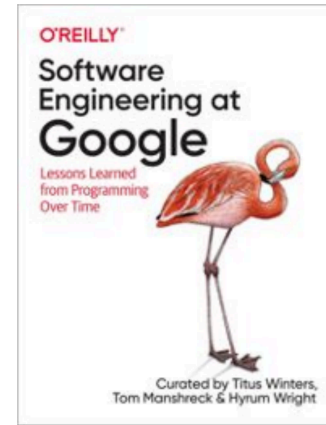
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## NOTES

The core idea is catching design flaws while they're still cheap to fix, before any implementation work has been done.

# Design documents

- The *best* design docs suggest design goals, and cover alternative designs, documenting the strengths and weaknesses of each.
- The *worst* design docs accidentally embed ambiguities, which cause implementers to develop contradictory solutions that the customer doesn't want.



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## NOTES

Ambiguity is the real enemy here: a design that reads clearly to the person who wrote it can still leave room for two engineers to build contradictory things, entirely independently, without either of them realizing it until integration.

# Companies using an RFC-like engineering planning process\*

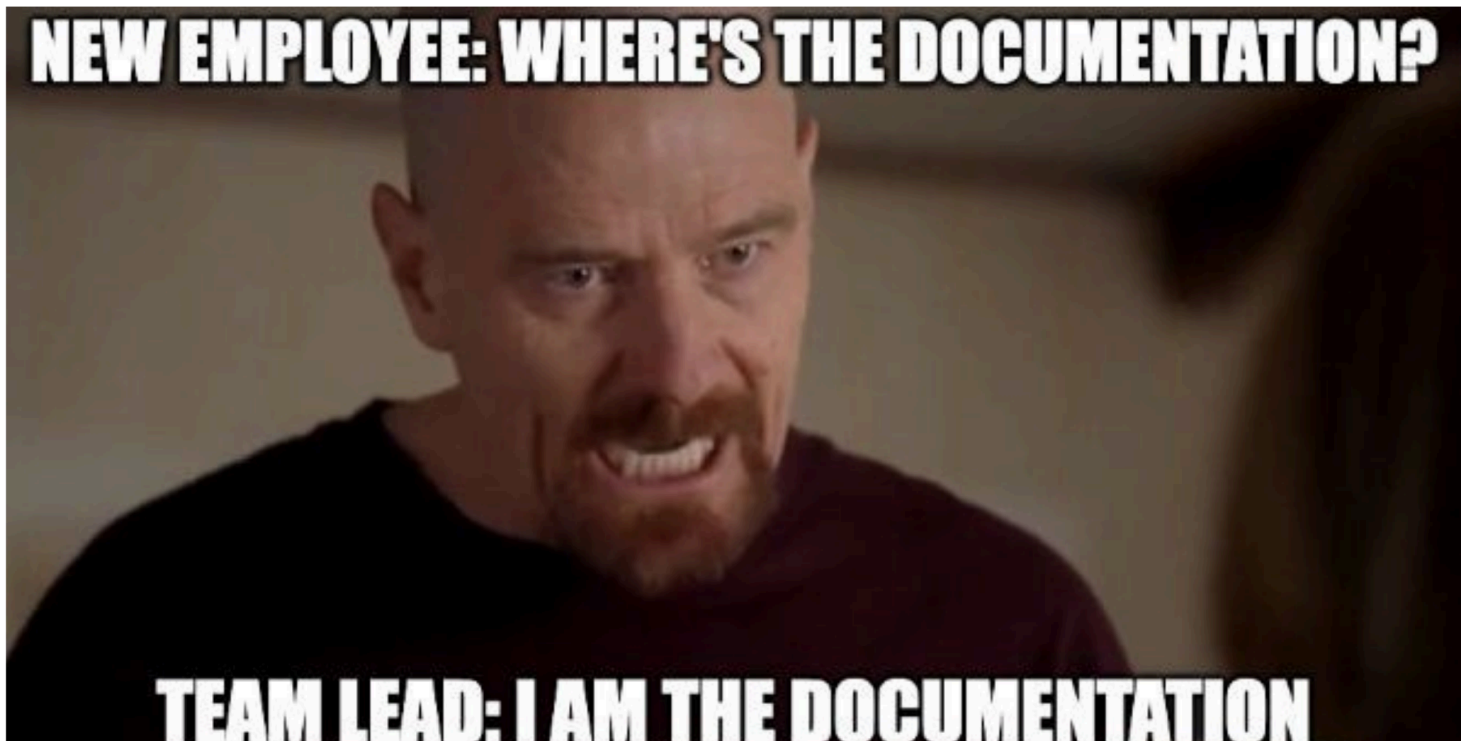
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pragmaticengineer.com

## NOTES

This isn't a niche practice; it's standard across serious engineering orgs of every size, from five-person startups to the largest tech companies.

# Why is this important?



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## NOTES

The joke lands because it's true: tribal knowledge that lives only in one person's head is a single point of failure for the whole team, and design docs are one of the cheapest ways to get it out of someone's head and onto paper.

This relates to the bus factor: the number of people who could suddenly become unavailable before the project gets into serious trouble. If only one person understands an important part of the system, the bus factor is one.

The name is intentionally dramatic. A friendlier version is the vacation factor: can someone take a vacation without the rest of the team getting stuck or repeatedly calling them?

# Common parts/templates

1. Metadata: *version, date, authors*
2. Executive Summary: *problem being solved, project mission*
3. Stakeholders (and non-stakeholders)
4. Scenarios / User Stories
5. User Experience
6. High-level Requirements: Functional
  - Global Requirements: Quality, Security, Privacy, Ethics
7. Features and Operations
8. Design Considerations and Tradeoffs
9. Non-Goals
10. Roadmap / Timeline
11. Open Issues

```
/*
 * Dear maintainer:
 *
 * Once you are done trying to 'optimize' this module
 * and have realized what a terrible mistake that was,
 * please increment the following counter as a warning
 * to the next person:
 *
 * total hours wasted here = 342
 */
```

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## NOTES

No template here is mandatory; treat this as a checklist of things a good design doc usually covers, not a form to fill in mechanically. Skip sections that genuinely don't apply.

# Examples: RFCs

## Requests for Comments

<https://datatracker.ietf.org/doc/html/rfc8925>

<https://rust-lang.github.io/rfcs/0019-opt-in-builtin-traits.html>



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### NOTES

Sourcegraph used to publish its design documents and RFC process publicly. They were excellent examples because students could see real proposals, alternatives, tradeoffs, comments, and decisions from an actual software company, rather than only looking at an empty template. Unfortunately, those public design docs are no longer available at the original links. This is also a small lesson about documentation: external resources and links can disappear, so important examples may need to be preserved or archived.

# When to use an RFC:



- You want to frame a problem and propose a solution.
- You want thoughtful feedback from team members on our globally-distributed remote team.
- You want to surface an idea, tension, or feedback.
- You want to define a project or design brief to drive project collaboration.
- You need to surface and communicate around a highly cross-functional decision with our formal decision-making process.

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## NOTES

We can still use Sourcegraph to discuss the RFC process and what made those documents valuable, even if we can no longer inspect the original collection live.

The common thread across all of these motivations is that you need real feedback from other people, not just a decision you've already made and want to announce.

# Don't use an RFC when



- You want to discuss personal or sensitive topics one-on-one with another team member.
- You want to decide to change something where you are the decider. In the vast majority of cases, creating an RFC to explain yourself will be overkill. RFCs should only be used if a decision explicitly requires one of the bullets in the previous page.

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## NOTES

If you already know what you're going to do and just want to announce it, an RFC is the wrong tool: that's what a design doc or a plain announcement is for. RFCs cost everyone's time; reserve them for when you genuinely need input.

- **WIP:** The author is still drafting the RFC and it's not ready for review.
- **Review:** The Review label is used when the RFC is ready for comments and feedback.
- **Approved:** When the RFC is for the purpose of making a decision, the Approved label indicates that the decision has been made.
- **Implemented:** When the RFC is for the purpose of making a decision, the Implemented label indicates that the RFC's proposal has been implemented.
- **Closed:** When the RFC is for collaboration or discussion but not necessarily to make a decision or propose a specific outcome that will eventually become Implemented, the Closed label indicates that the RFC is no longer an active collaborative artifact.
- **Abandoned:** When the RFC is for the purpose of making a decision, and there are no plans to move forward with the RFC's proposal, the Abandoned label indicates that the RFC has been purposefully set aside.

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## NOTES

These labels exist so anyone browsing the RFC list can tell at a glance whether something is still open for input, already decided, or no longer active, without having to read the whole document. However, there is no universal set of RFC labels. As we saw when browsing different repositories in class, teams use different names and workflows, and some apply their labels inconsistently. Consistency is key: either use status labels systematically or do not rely on them. Applying them only sometimes makes the repository harder to understand and reduces their value.

# Design Docs in the Wild

- Many Design Docs are publicly available
  - [Design Doc Library](#)
- Let's take a look at [one](#)!

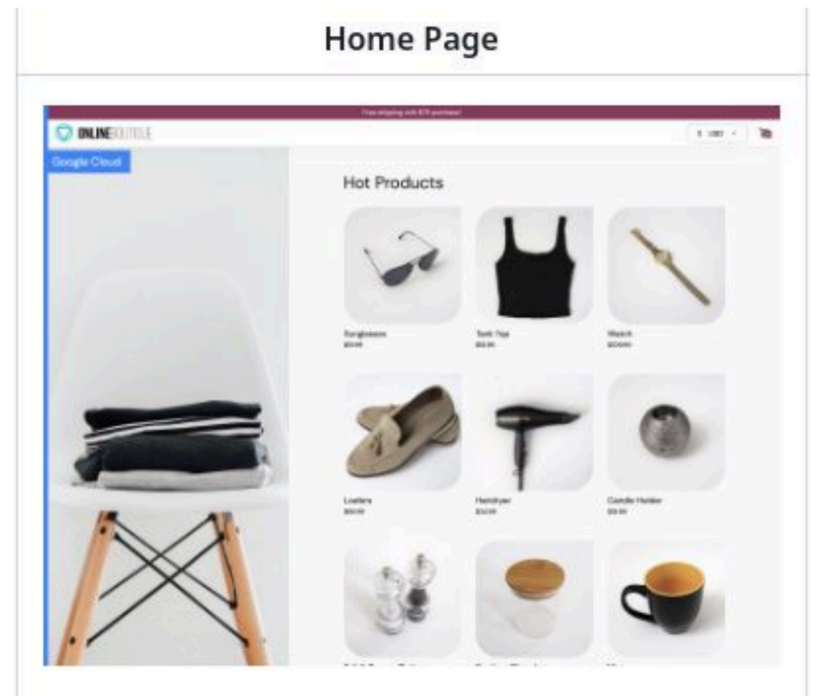
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## NOTES

Notice how much of a real RFC is spent on tradeoffs and alternatives considered, not just the final decision; that's usually the most valuable part for a reader trying to understand why a choice was made.

# Exercise

- 3 Proposed Features:
  - Add Payment Method
  - Personalized Promotions
  - Add Android Support



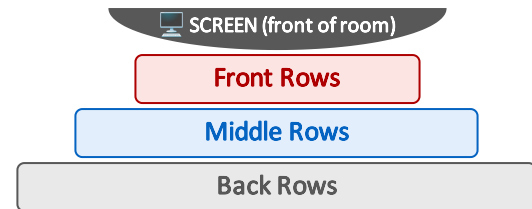
Try the demo: [bit.ly/313-boutique](https://bit.ly/313-boutique)

# Time to write our own design docs!

- Divide up into 3 sections, based on where you're sitting

(NOTE: you should be signed in w/Andrew to Google)

- Your mission:
  - Write a short design doc (RFC-style) for your assigned feature in the Online Boutique
  - Collaborate in real time on a shared Google Doc
  - Everyone should contribute ideas, questions, or text.
  - Build on others' work. Use comments instead of deleting their contributions.
- Be creative. Examples, sketches, diagrams, and emojis are welcome.
- Keep all contributions constructive and professional.



Add Payment Method



[bit.ly/f26-doc-payment](https://bit.ly/f26-doc-payment)

Personalized  
Promotions



[bit.ly/f26-doc-promotions](https://bit.ly/f26-doc-promotions)

Add Android Support



[bit.ly/f26-doc-android](https://bit.ly/f26-doc-android)

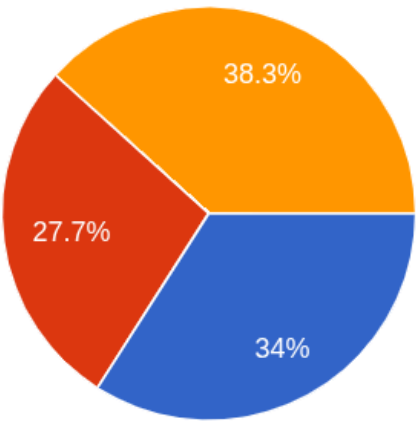
Vote



# Which design doc is the best?

Which design doc is the best?

47 responses



- Add Android Support
- Add Payment Methods
- Personalized Promotions

NOTES

Results from Fall 2026! Congratulations to the winners!